# Exit Ticket · reflect and retrieve

## Exit Ticket

A hash-table function based on adding character codes is suitable for protecting real passwords.

## Use key vocabulary from memory

1. Explain A table using a hash function to locate data from a key. in your own words.

2. Explain A repeatable mapping from key to bucket index. in your own words.

3. Explain A location in the hash table. in your own words.

## Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.



RHS COMPUTING / OCR H446 / LESSON 033 DRAFT · TEACHER REVIEW

# Do Now · retrieval

## C# Login Project · Code a Hash Table

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

### Q1

A login check uses validUser && validPassword . What is the result when validUser is true and validPassword is false? Explain.

### Q2

A D-type flip-flop currently stores Q = 1. D changes to 0 but there is no triggering clock edge. What is Q?

### Q3

A queue receives A, B and C in that order. Remove one item, then enqueue D. List the remaining items from front to rear.

### Q4

A five-bucket hash table uses h(k) = k MOD 5 with linear probing. Insert keys 7, 12 and 17 in that order into an empty table. State each final index.

# New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

## Core idea 1

The problem: more than one user — A single username comparison is not scalable. We need a way to store and find many usernames.

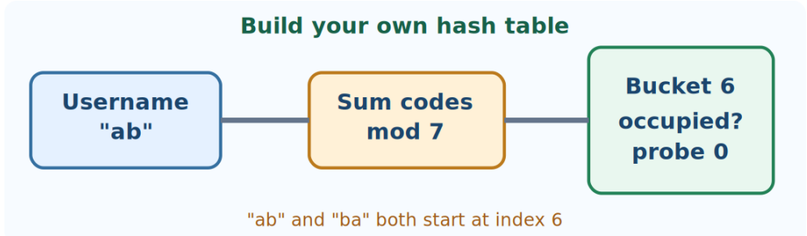
## Core idea 2

What is a hash function? — Convert the username to a number, then use % tableSize to choose a bucket index. Our small function is deliberately simple for learning.

## Core idea 3

What if two names choose one bucket? — That is a collision. Use linear probing: try the next bucket, wrapping to the beginning when necessary.

**Key visual · complete concept**




## Annotate the key visual

Explain which labels or features show the main process.





# You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 2

SYNOPTIC MCQ BANK
Write MCQs on linked lists, directed/undirected graphs, stacks, queues, trees, binary search trees and hash tables. Make a six-question draft; each needs four options and a written explanation.

Check that you have used the technical terms and explained your decisions.

# You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 1

CODING 1 · INSERT
Create a seven-bucket string array, implement Hash(), and insert at least four test usernames. Print each chosen bucket index.

Check that you have used the technical terms and explained your decisions.





# New Learning · applying the ideas

## Core idea 4

Hash-table lookup versus password hashing — The hash-table function locates an account. It does NOT securely protect a password. Use invented demo passwords only.

## Core idea 5

MCQ revision — Write four precise MCQs on linked lists, stacks, queues and trees; extend with BST and hash-table MCQs if time allows.

## Vocabulary · use the correct technical terms

A table using a hash function to locate data from a key. — A table using a hash function to locate data from a key.

A repeatable mapping from key to bucket index. — A repeatable mapping from key to bucket index.

A location in the hash table. — A location in the hash table.

Two keys map to the same bucket. — Two keys map to the same bucket.

## Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

# I Do · worked example

## Problem

Build a real hash-table index

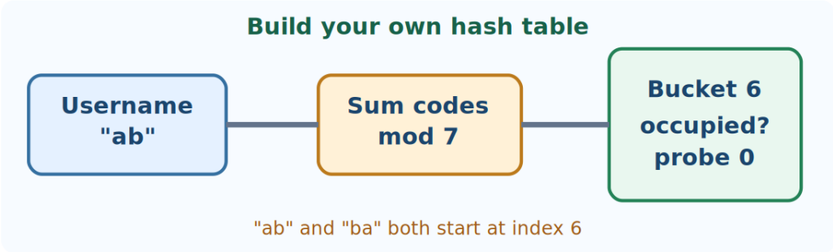
## Follow the worked method

1. Create the buckets — CODE:
```
const int SIZE = 7;
string?[] usernames = new string?[SIZE];
```

2. Write the index function — CODE:
```
static int Hash(string key, int size)
{
int total = 0;
foreach (char letter in key)
total += (int)letter;
return total % size;
}
```

3. Explain the limitation — CODE:
```
// Hash("ab", 7) == Hash("ba", 7)
// Same total -> collision.
// This is an educational hash function,
// NOT a secure password hash.
```

**Source model · final illustrated step**




## Why this method works

Identify the decision, rule or invariant that makes the method correct.





# We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

## Guided problem · TRY

Explain your solution first
What happens when a bucket is already occupied?

## DISCUSS

Compare how you reached the answer. Where could someone make an error?

## IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.